

DHRUV BHAVESH PAREKH

Graduate Student | Electrical & Computer Engineering | University of Texas at Austin

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EDUCATION

University of Texas at Austin

MSE | Electrical & Computer Engineering | Power Electronics and Power Systems | GPA - 4.0/4.0

Aug'24 - May'26

Indian Institute of Technology Gandhinagar

B.Tech | Electrical Engineering | GPA - 9.26/10 | Institute Gold Medal

Aug'20 - May'24

WORK EXPERIENCE

Texas Instruments

Systems Engineering Intern | ASM - AUTO

Jun'25 - Aug'25

- Motor-control algorithm design, validation in MATLAB Simulink and implementation on F29 series micro-controller.
- Test-bed design for comparative evaluation versus SVPWM for various test conditions.

University of Texas at Austin

Graduate Research Assistant | Prof. Brian Johnson

Jan'25 - Present

- Designing a GFM verification testbed experiment to run a frequency sweep test on any device to verify functioning.

MITACS Globalink | University of Waterloo

Research Intern | Prof. Sahar Azad

May'23 - Jul'23

- Designed & implemented various control loops for **Voltage-Sourced-Converters based Grid Forming Inverters** in PSCAD
- Simulated and ran comparative analysis of **droop control methods** based on transient time and the steady state error

Indian Institute of Technology Bombay

Research Intern | Prof. Zakir Rather

May'22 - Jul'22

- Collaborated on a pilot project to develop and deploy **Street-Pole based Charging for Electric Vehicles** in IIT Bombay
- Designed and developed an Arduino based circuit to ensure Pilot and Proximity checks for a J-1772 based EVSE

PROJECTS

2 stage Miller Compensated OTA Design

Course Project | Prof. Yaoyao Jia | University of Texas at Austin

Aug'25 - Present

- Complete design of a 2-stage OTA to meet requirements pertaining to output swing, DC gain and transient performance.
- Simulation of design in Cadence for verification under various step changes in input.

Model Predictive Control of a Boost Converter

Course Project | Prof. Sandeep Chinchali | University of Texas at Austin

Aug'25 - Dec'25

- Comparative analysis of LQR, iLQR, and QP in MPC against normal PID Control by simulation in PLECS.

Design for a Boost Converter

Course Project | Prof. Brian Johnson, Prof. Alex Hanson | University of Texas at Austin

Aug'24 - Present

- Completely designed a boost converter circuit that involves device selections, inductor design, and PCB layout.
- Experimentally verified the design, achieved one of the highest efficiency in class (98 Percent), minimizing converter size.
- Implementation of Peak-Curve control and tested dynamic performance under multiple load changes.

Comparison of Different Modulation Techniques for an Inverter

Research Project | Prof. Ragavan K | IIT Gandhinagar

Jan'24 - Apr'24

- Implemented various PWM Techniques on **STM** aimed at reducing **Total Harmonic Distortion** in Inverters
- Conducted Experimental and simulation based comparison to determine most effective PWM Technique

TOOLS & FRAMEWORKS

PSCAD, MATLAB, Simulink, LTSpice, Arduino, Python, Verilog, LabVIEW, KiCad, PLECS, CCS, STM32, TCAD, Cadence.